Class 3

Sabse pehle kya bana?

ENIAC (1940s)-Electronic Numerical Integrator and Computer

- **CPU (processing unit)**
- **Memory**
- **Input/output devices**

Problem ye thi:Computer ko kaise bataye ki kya kaam karna hai?

Direct Wiring Programming

Wires aur switches use karke program set karte the

Matlab literally machine ke andar wiring change karni padti thi

Logon ne bola: "Yaar ye wiring bahut tough hai"

Punch cards use karo

Code ko cards me holes ke form me store karo

Machine me daalo → program run

Machine Language kya hoti hai?

CPU sirf ON / OFF signals samajhta hai

- **ON = 1**
- **OFF = 0**

Binary (0/1) likhna mushkil tha

Phir Assembly Language aayi

👉 To logon ne short commands banayi:
MOV, ADD, SUB
👉 Ye assembly language thi
Fir assembler convert karta tha machine code me

1950s–60s

Early Systems (IBM etc.)

IBM ne early computers aur OS banaye
Example: **GM-NAA I/O, OS/360**

Closed source

1969s

Bell Labs (AT&T)

Ye modern OS ka base bana

License cost: thousands of dollars ($1000 – $10000+)
Saath me:
Expensive hardware (mainframes/workstations)
Maintenance cost

Normal aadmi afford hi nahi kar sakta tha 😅
Sirf:
Research labs
Big companies
Universities

Closed source

1980s

Microsoft

MS-DOS,Baad me Windows (Windows 95, 98)

MS-DOS (1980s)
Approx cost: $40 – $100
Mostly PC ke saath bundle aata tha
👉 Matlab:
Individual log bhi afford kar sakte the

Windows 98: ~$100–$200
Home users ke liye design kiya gaya
GUI (mouse + windows) ne easy bana diya

open source

1991s

Linus Torvalds

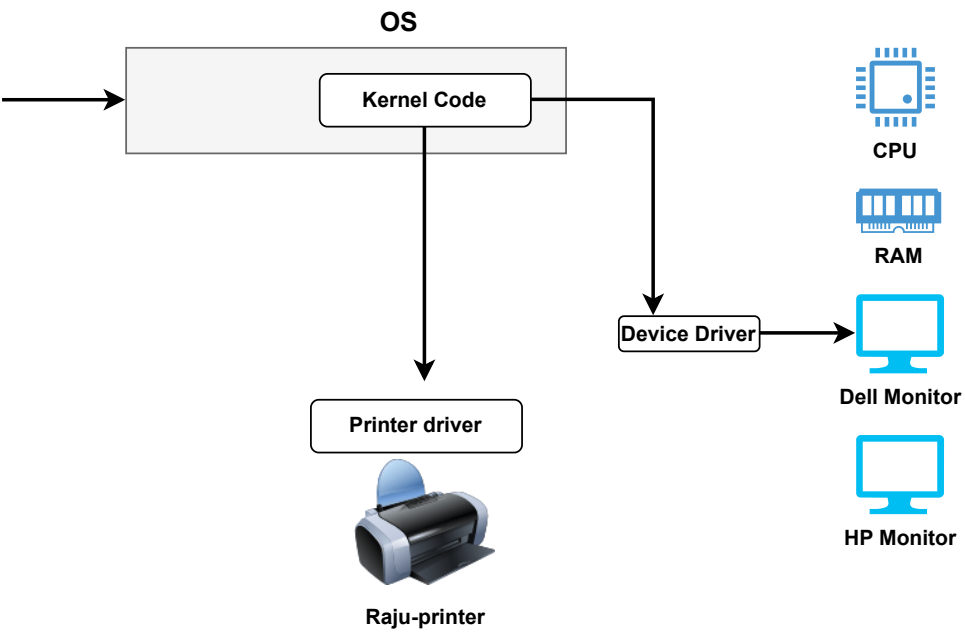
Linux Kernel

Inspired by UNIX,But **UNIX ka copy nahi hai**

Operating System (OS) = pura system

Kernel = OS ka core part (heart ❤️)

OS=GUI (Windows screen, icons)+Applications (notepad, browser)+File system+Kernel



Linux kernel (Open source)

- **Linux kernel is free**
- **Linux kernel is secure**

Closed source

Code kahi nhi dikhenga

Open source tools

- **Terraform**
- **Kubernetes**
- **Linux**

Open source

- **Code Git hub pr milega**
- **bahut saare log milkar isko free & secure bna rhe hain**
- **365 days/24*7 suport**
- **Performance**

Linux based Operating System

- **Ubuntu Linux**
- **Kali linux**
- **Parrot OS**
- **Android**

Linux/Mac/Linux

GUI Mode(Graphical User Interface)
Click click karke kaam karte hai
Bulky

CLI Mode
Command line interface
command se maja lo
Very lightwieght